7th Order Butterworth Analog Filter

Applications

- Anti-aliasing filter for ADCs
- Dynamic characterization of different sensors
- Analog Front-End filtering
- Demodulation for High-RPM speed sensors
- Vibrational analysis and acoustic
- Low-frequency ultrasound acquisition

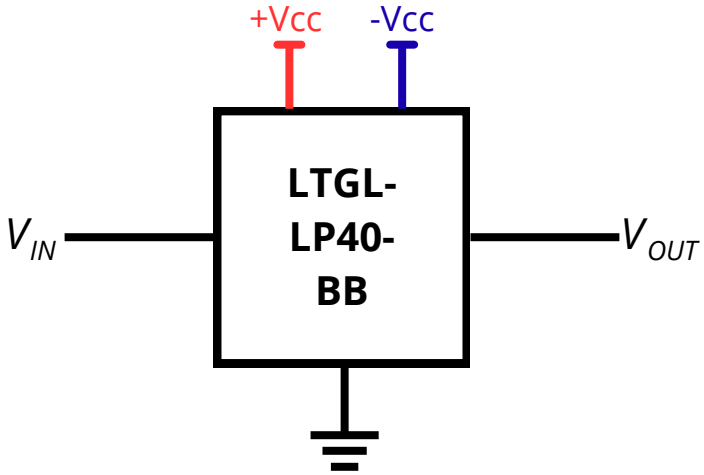
Description

The device **LTGL-LP40-BB** is a **seventh-order Butterworth low-pass filter** designed for generic analog signal conditioning and frequency selection applications.
The Butterworth approximation has been adopted in order to achieve a maximally flat magnitude response within the passband, ensuring **low signal distortion and smooth frequency behavior**.

The filter is characterized by a -3 dB cutoff frequency (relative to the DC gain) at 40 kHz, while providing a stopband attenuation of -13 dB at the frequency of 61 kHz.
The circuit has been designed with an overall DC gain = 5, providing stable operation and consistent response over the specified operating range.
The seventh-order topology enables a relatively sharp transition between passband and stopband regions while preserving the monotonic response typical of Butterworth filters.

This filter is intended as a **general-purpose solution** suitable for medium-frequency analog processing systems, data acquisition front-ends, and signal filtering applications requiring smooth passband characteristics and moderate stopband rejection.

Block Diagram



Specifications and Ratings

- ## Absolute Maximum Ratings

Supply Voltage (±Vcc)	±18 V
Input Voltage (±Vin)	±15 V
Operating Temperature Range	-55 °C to 125 °C

- ## Testing Operating Conditions

Supply Voltage (±Vcc)	±10 V
Input Voltage (±Vin)	±500 mV
Operating Temperature	Room Temperature (20 °C)

- ## Ideal Electrical Characteristics

Filter Type	Butterworth
Cutoff Frequency	40 kHz
Stopband Frequency	61 kHz
Maximum Band Pass Attenuation	2 dB
Minimun Cutoff Band Attenuation	20 dB
DC Gain	5

Filter design

- **Filter order**

With the requirements of the filter

$$f_p = 40\ kHz \rightarrow\ \omega_p = 2\pi\ f_p \approx 251.33\ krad/s$$
$$f_s = 61\ kHz \rightarrow\ \omega_s = 2\pi\ f_s \approx 383.27\ krad/s$$
$$A_{MAX} = 2\ dB$$
$$A_{min} = 20\ dB$$
$$K = \ DC_{gain} = 5$$

it can be found that:

$$\varepsilon = \sqrt{10^{\frac{A_{MAX}}{10}} - 1} = 0.765$$

The formula for the attenuation, to find the filter order, is:

$$A(\omega_s) = 10 \log_{10}\left(1 + \varepsilon^2 \left(\frac{\omega_s}{\omega_p}\right)^{2n}\right)$$

Where the attenuation $A(\omega_s)$ must be calculated by iterations of n:

$n = 4$:

$$A(\omega_s, n = 4) \approx 12.58\ dB\ (NO)$$

$n = 5$:

$$A(\omega_s, n = 5) \approx 16.40\ dB\ (NO)$$

$n = 6$:

$$A(\omega_s, n = 6) \approx 19.20\ dB\ (NO)$$

$n = 7$:

$$A(\omega_s, n = 7) \approx 23.25\ dB > A_{min} = 20\ dB\ (YES) \rightarrow \boldsymbol{n = 7}$$

This can also be verified with the closed formula:

$$n \geq \frac{\log_{10}\left(\frac{10^2 - 1}{10^{0.2} - 1}\right)}{2 \log_{10}\left(\frac{61}{40}\right)} \approx 6.08\ \rightarrow \boldsymbol{n = 7}$$

- **Poles positioning**

Once the filter order has been determined, the pole locations need to be found, the formula for a generic pole is:

$$S_k = \omega_p e^{j\frac{\pi}{2}\left(1+\frac{2k+1}{n}\right)}$$

In this case $k = 0, \dots, 6$, which results in three pairs of complex conjugate poles and a real negative pole:

$$S_{0,6} = \omega_p e^{\pm j\frac{4\pi}{7}} = \omega_p \left(\cos\left(\frac{4\pi}{7}\right) \pm j\, sin\left(\frac{4\pi}{7}\right)\right) \rightarrow \theta_{0,6} = \pm\frac{4\pi}{7}$$

$$(s - s_0)(s - s_6) = s^2 - s\omega_p(-0.223 + j0.975) + -s\omega_p(-0.223 - j0.975) + s_0 s_6$$
$$= s^2 + 0.445\, s\omega_p + {\omega_p}^2$$

$$S_{1,5} = \omega_p e^{\pm j\frac{5\pi}{7}} = \omega_p \left(\cos\left(\frac{5\pi}{7}\right) \pm j\, sin\left(\frac{5\pi}{7}\right)\right) \rightarrow \theta_{1,5} = \pm\frac{5\pi}{7}$$

$$(s - s_1)(s - s_5) = s^2 - s\omega_p(-0.623 + j0.782) + -s\omega_p(-0.623 - j0.782) + s_1 s_5$$
$$= s^2 + 1.246\, s\omega_p + {\omega_p}^2$$

$$S_{2,4} = \omega_p e^{\pm j\frac{6\pi}{7}} = \omega_p \left(\cos\left(\frac{6\pi}{7}\right) \pm j\, sin\left(\frac{6\pi}{7}\right)\right) \rightarrow \theta_{2,4} = \pm\frac{6\pi}{7}$$

$$(s - s_2)(s - s_4) = s^2 - s\omega_s(-0.9 + j0.434) + -s\omega_s(-0.9 - j0.434) + s_2 s_4$$
$$= s^2 + 1.8\, s\omega_p + {\omega_p}^2$$

$$S_3 = \omega_p e^{-j\pi} = -\omega_p \rightarrow (s + \omega_p)$$

So, the transfer function of the filter is:

$$H(s) = \frac{5}{(s + \omega_p)(s^2 + 0.445\, s\omega_p + {\omega_p}^2)(s^2 + 1.246\, s\omega_p + {\omega_p}^2)(s^2 + 1.8\, s\omega_p + {\omega_p}^2)}$$

- **Quality factor Q and Denominators**

The generic quality factor Q_i and denominator $D_i(s)$ are:

$$Q_i = \frac{1}{2|\cos(\theta_i)|}$$

$$D_i(s) = s^2 + \frac{\omega_p}{Q_i}s + {\omega_p}^2$$

For this filter:

$$Q_{4°stage}(s_{0,6}) = \frac{1}{2|\cos(\theta_{0,6})|} \approx 2.247 \rightarrow D_{4°stage}(s) = s^2 + 1.1620 \cdot 10^5 s + 6.8184 \cdot 10^{10}$$

$$Q_{3°stage}(s_{1,5}) = \frac{1}{2|\cos(\theta_{1,5})|} \approx 0.802 \rightarrow D_{3°stage}(s) = s^2 + 3.2562 \cdot 10^5 s + 6.8184 \cdot 10^{10}$$

$$Q_{2°stage}(s_{2,4}) = \frac{1}{2|\cos(\theta_{2,4})|} \approx 0.555 \rightarrow D_{2°stage}(s) = s^2 + 47051 \cdot 10^5 s + 6.8184 \cdot 10^{10}$$

- **First Stage: Inverting single-pole stage with gain K=5**

  For a single inverting amplifier stage with nominal gain (K = 5), the voltage gain is given by:

$$K = -\left(\frac{R_2}{R_1}\right)$$

  Since a single $20\ k\Omega$ resistor was not available, the feedback resistor was implemented using two $10\ k\Omega$ resistors in series:

$$R_{2.1} = 10\ k\Omega$$
$$R_{2.2} = 10\ k\Omega$$
$$R_2 = R_{2.1} + R_{2.2} = 20\ k\Omega$$

  The resistor R_1 is calculated as:

$$R_1 = R_2 \div K = 20\ k\Omega \div 5 = 4\ k\Omega$$

  A standard laboratory value was used:

$$R_1 = 3.9\ k\Omega$$

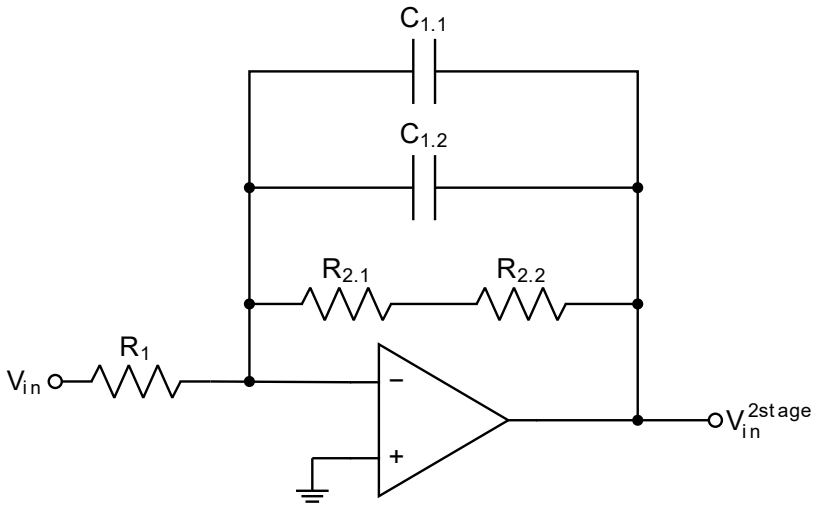
  The capacitor value for the first stage was calculated by setting the pole frequency to $f_p = 40\ kHz$, corresponding to $\omega_p = 2\pi f_p$.

  Using the equivalent resistance $R_2 = 20\ k\Omega$, the capacitance is:

$$C = \frac{1}{20k\Omega \cdot 2\pi \cdot 40kHz} \approx 200pF$$

  A standard value of 200 pF was therefore selected, implemented as two capacitors in parallel.

$$C_{2.1} = 100pF$$
$$C_{2.2} = 100pF$$
$$C_2 = C_{1.1} \,||\, C_{1.2} = 200pF$$



First stage circuit schematic

- **Second stage: Sallen-Key SMART with Q = 0.555 and K = 1**

  The second stage was designed using a Sallen-Key filter topology starting with the selection of an arbitrary capacitance:

  $$C_3 = 330\ pF$$

  Using the design constraint:

  $$R_3 = \frac{2 \cdot Q}{C_3 \cdot \omega_p}$$

  Substituting the values:

  $$R_3 = \frac{2 \cdot 0{,}555}{330\ pF \cdot 2.51 \cdot 10^5} = \frac{1.11}{330 \cdot 10^{-12} \cdot 2.51 \cdot 10^5}$$
  $$= \frac{1.11}{8.28 \cdot 10^{-5}} \approx 13.41\ k\Omega$$

  Standard laboratory value selected:

  $$R_3 \approx 13.3\ k\Omega$$

  Implemented as $R_{3.1} = 10\ k\Omega$ + $R_{3.2} = 3.3\ k\Omega$ in series.

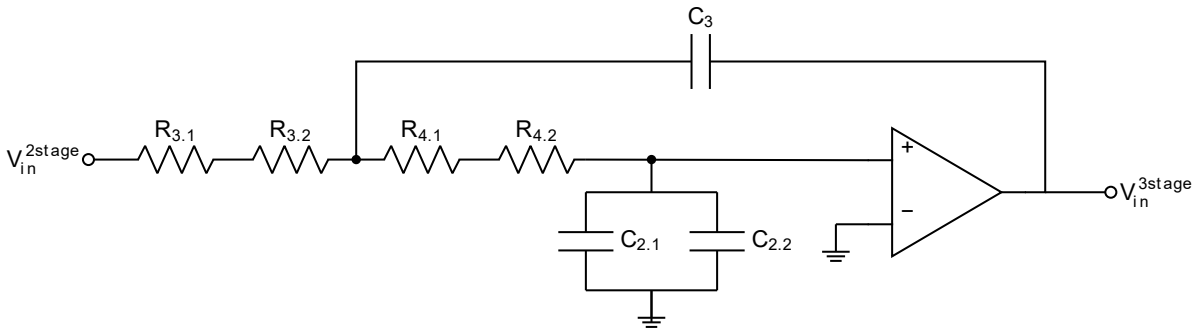
  Then, based on the design constraint: $R_3 = R_4 = 13.3\ k\Omega$

  Finally, we get the last capacitor's value:

  $$C_3 = \frac{C_2}{4 \cdot Q^2} = \frac{330\ pF}{4 \cdot 0.555^2} = \frac{330 \cdot 10^{-12}}{4 \cdot 0.308} \approx 268\ pF$$

  Commercial values were chosen: $C_{3.1} = 220\ pF$, $C_{3.2} = 47\ pF$ placed in parallel:

  $$C_{3.1} || C_{3.2} = 267\ pF$$



Second stage circuit schematic

- **Third stage: Sallen-Key smart with Q = 0.805 and K = 1**

The design started by selecting an arbitrary standard capacitor value:

$$C_5 = 470\ pF$$

Using the design constraint:

$$R_5 = \frac{2 \cdot Q}{C_5 \cdot \omega_p}$$

$$R_5 = \frac{2 \cdot 0{,}805}{470\ pF \cdot 2.51 \cdot 10^5} = \frac{1.61}{470 \cdot 10^{-12} \cdot 2.51 \cdot 10^5} = \frac{1.61}{1.179 \cdot 10^{-4}} \approx 13.65\ k\Omega$$

Standard laboratory value selected:

$$R_5 \approx 13.3\ k\Omega$$

Implemented as $R_{5.1} + R_{5.2} = 10\ k\Omega + 3.3\ k\Omega$

In order to achieve a unity gain, the resistors must satisfy the condition $R_5 = R_6$, therefore:

$$R_6 = 13.3\ k\Omega$$

The value of $C_4$was determined by applying the following capacitive design constraint:

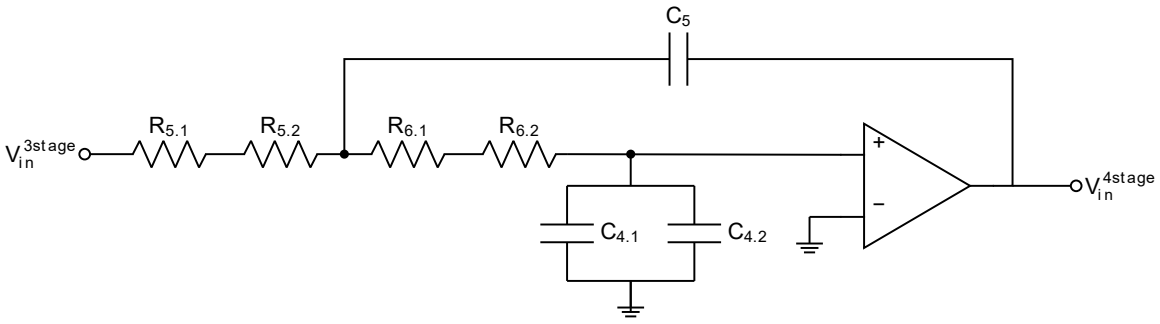
$$C_4 = \frac{C_5}{4 \cdot Q^2}$$

$$C_4 = \frac{470pF}{4 \cdot 0.805^2} = \frac{470 \cdot 10^{-12}}{4 \cdot 0.648} \approx 181\ \text{pF}$$

Standard laboratory value selected:

$$C_4 = 182\ pF$$

Implemented as $C_{4.1} || C_{4.2} = 100pF + 82pF$



Third stage circuit schematic

- **Fourth stage: Tow-Thomas with Q = 2.247 and K = 1**

  The last stage was designed using a Tow-Thomas filter topology, since the Quality Factor is very high.

  The first capacitance is arbitrary and commercial:

  $$C_6 = 220\ pF = C_7$$

  Then, knowing ω_p, we get the value:

  $$R_7 = \frac{1}{\omega_p C_6} = \frac{1}{220 \cdot 10^{-12} \cdot 2.51 \cdot 10^5} = \frac{1}{8.28 \cdot 10^{-5}} \approx 18{,}1\ k\Omega$$

  Standard laboratory value selected:

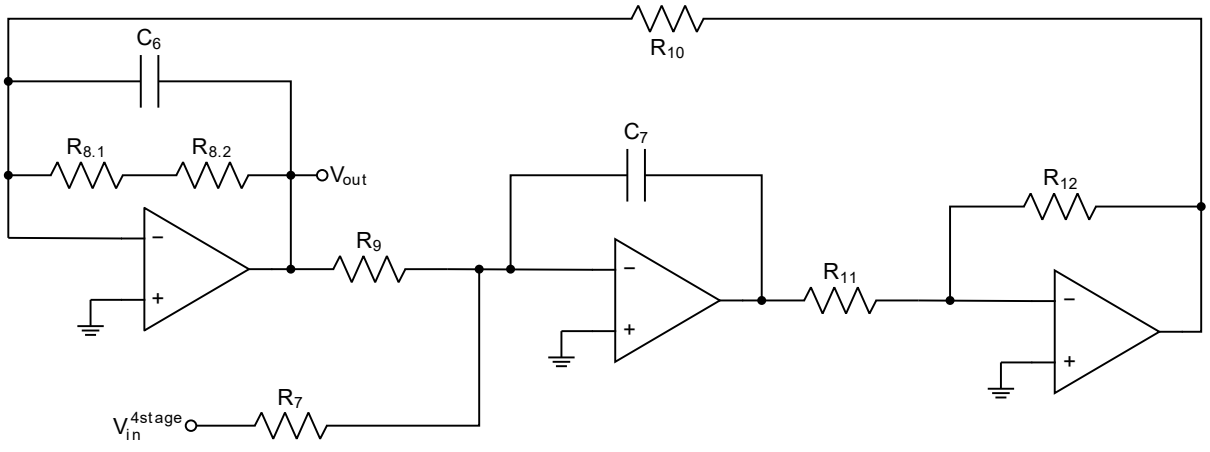
  $$R_7 \approx 18\ k\Omega = R_9 = R_{10}$$

  Then, based on the design constraint: $R_8 = Q \cdot R_7 = 2.247 \cdot 18\ k\Omega = 40.5\ k\Omega$

  Implemented with commercial values: $R_{8.1} = 18k\Omega$ and $R_{8.2} = 22k\Omega$.

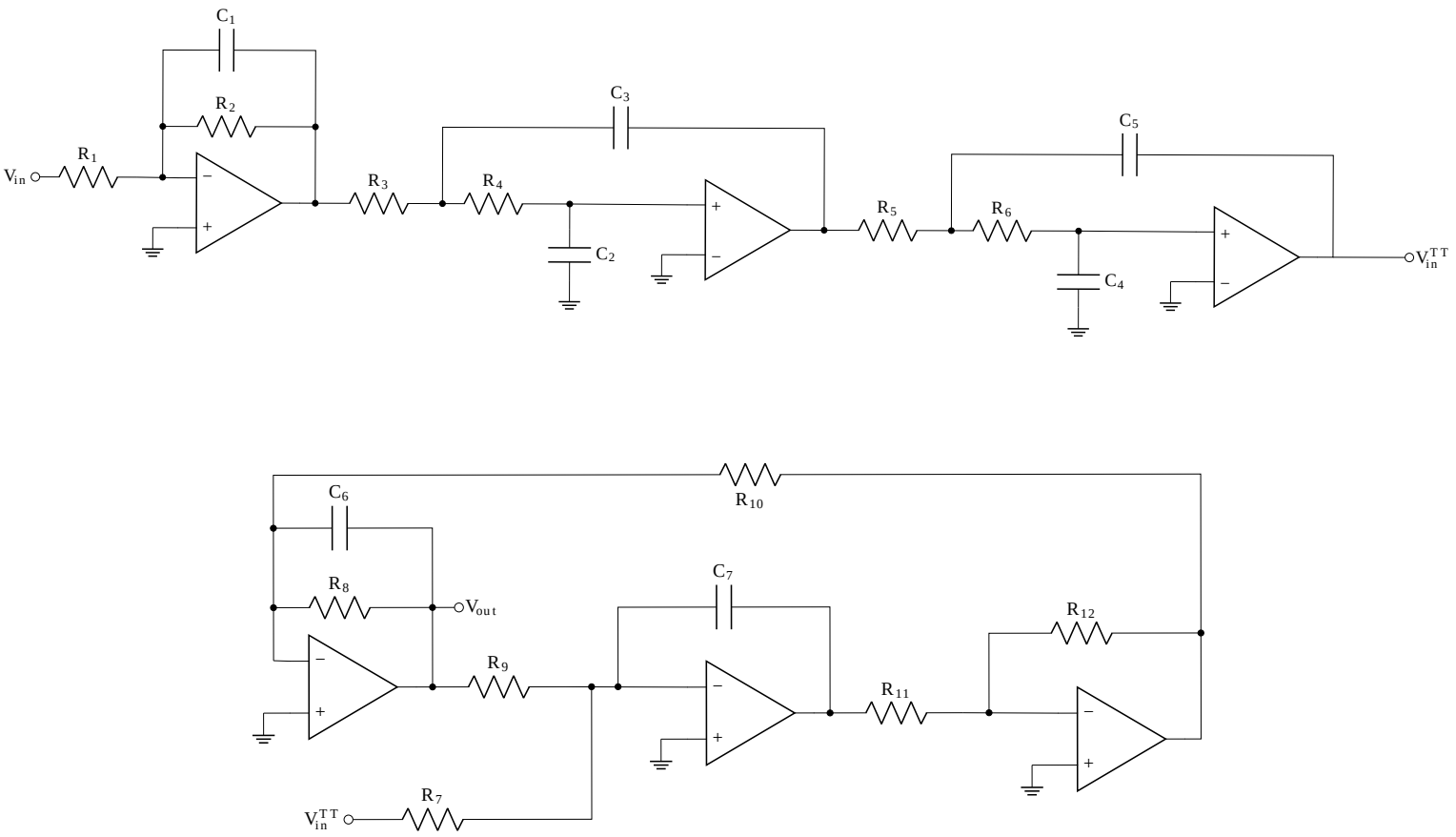
  Finally, to set the last step as an inverter with unit gain we selected:

  $$R_{11} = R_{12} = 10k\Omega$$



Fourth and last filter stage

- **Final Schematic**

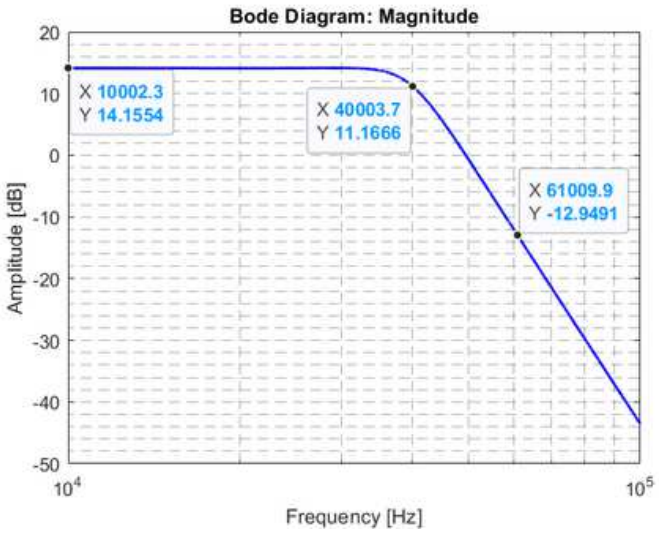


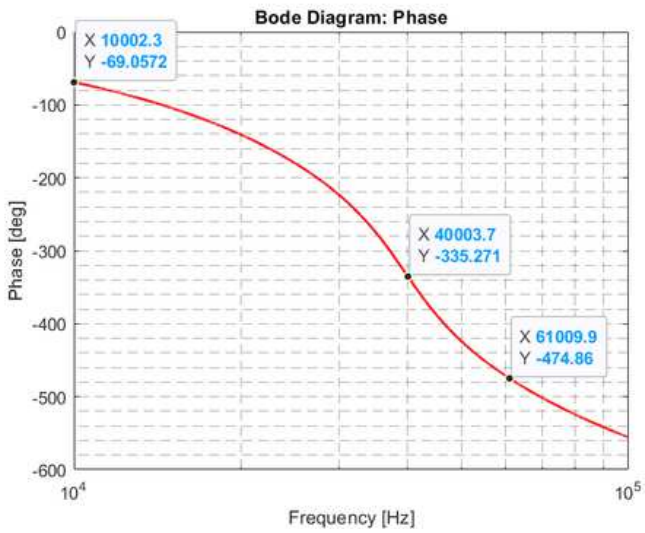
- **Commercial Op-Amps Used**

Stage	Op-Amp used	Description
First	LM741	Simulations showed the lower-grade op-amp was more sensitive to variations for this stages, for this reason we chose a trade-off solution
Second (S&K)		
Third (S&K)		
Fourth (TT)	TL084	Chosen for the most critical stage due to its good performance. Also, it's quad op-amp package, saves space

Simulations

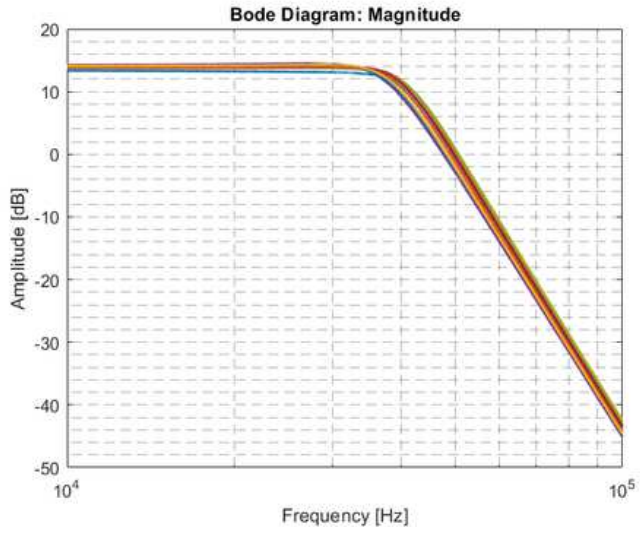
- **Ideal Bode Plots**

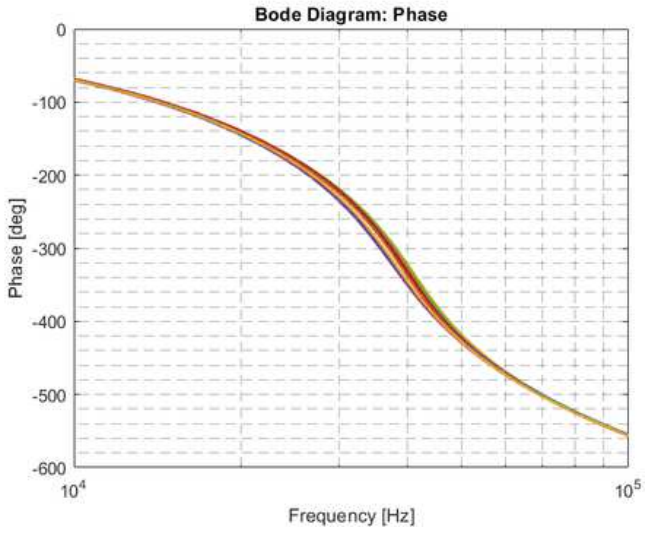




Bode Diagram Analysis: The ideal plots demonstrate the expected frequency response of the filter, clearly illustrating the passband gain and the subsequent attenuation slope. The phase transition is consistent with the filter's order, providing a precise baseline for the expected phase lag across the analyzed frequency spectrum.

- **Montecarlo Bode Plots**





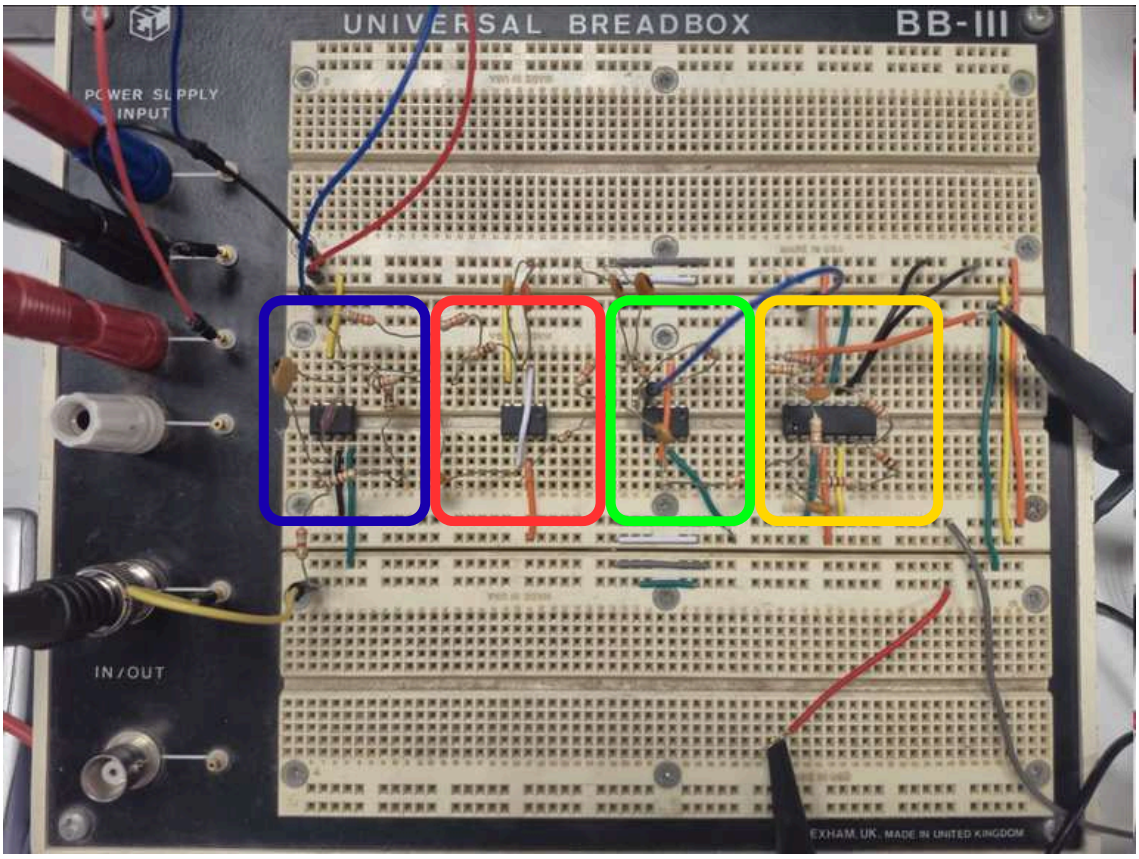
Monte Carlo Sensitivity Analysis: The Monte Carlo simulations, performed across 30 iterations, provide a robust evaluation of the filter's performance under parametric variations. The resulting plots effectively visualize the design's sensitivity, showing a tight clustering of responses that confirms high reliability and stability despite component tolerances.

- **Montecarlo Results**

Frequency	Minimum (Magnitude / Phase)	Maximum (Magnitude / Phase)
10 kHz	13.36 / -70.60	14.25 / -68.35
40k Hz	9.16 / -351.25	11.78 / -322.50
61 kHz	-14.93 / -476.94	-11.62 / -471.88

Test setup and equipment

- **Breadboard setup**



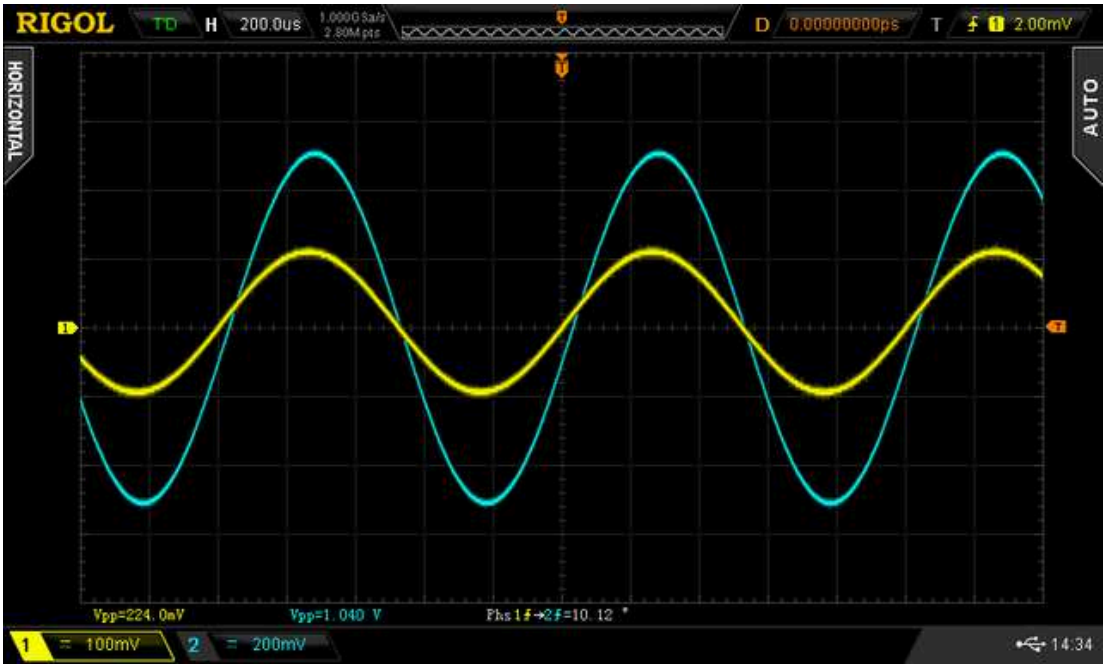
First Stage **Second Stage** **Third Stage** **Fourth Stage**

- **Test equipment**

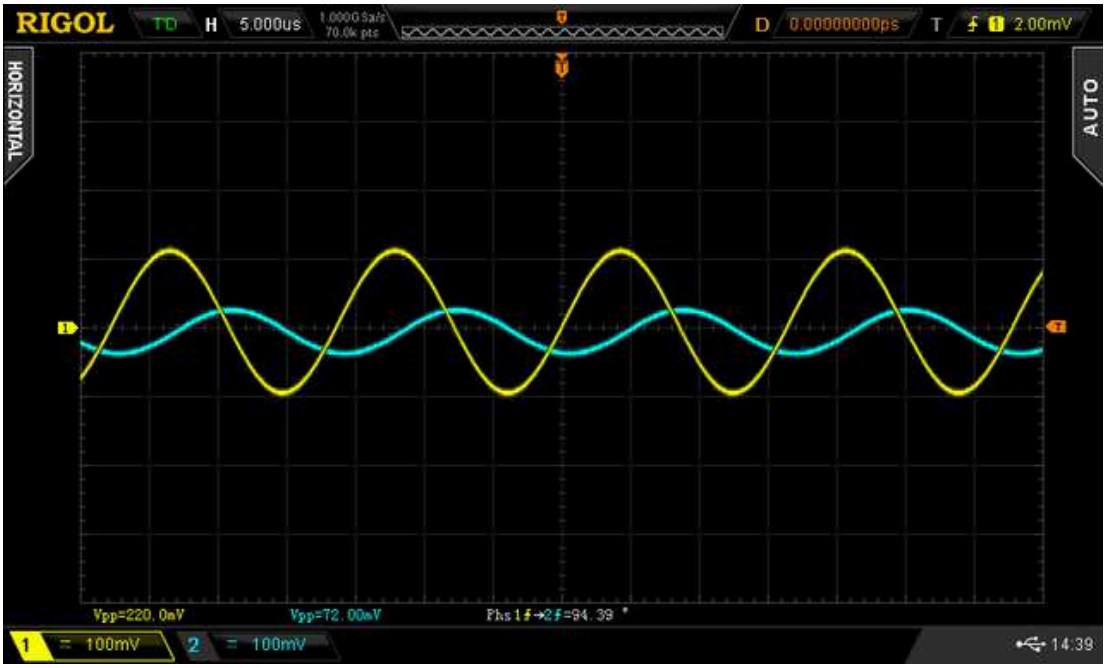
Multimeter	Keithley 2210
Oscilloscope	Rigol DS2302A
Function Generator	Agilent 33210A
Power Supply	Rigol DP831

Test results

- **Oscilloscope Pictures**

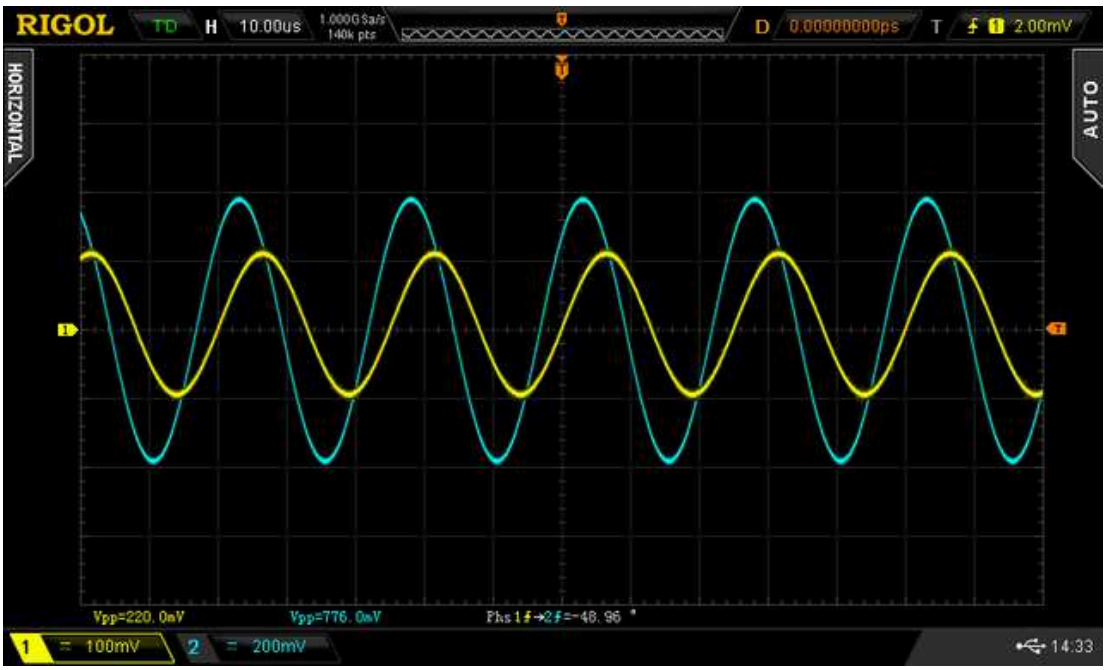


1 kHz Input: The input signal (blue) passes through the filter with minimal attenuation and negligible phase shift (10.12°), confirming that the filter operates correctly within its passband.



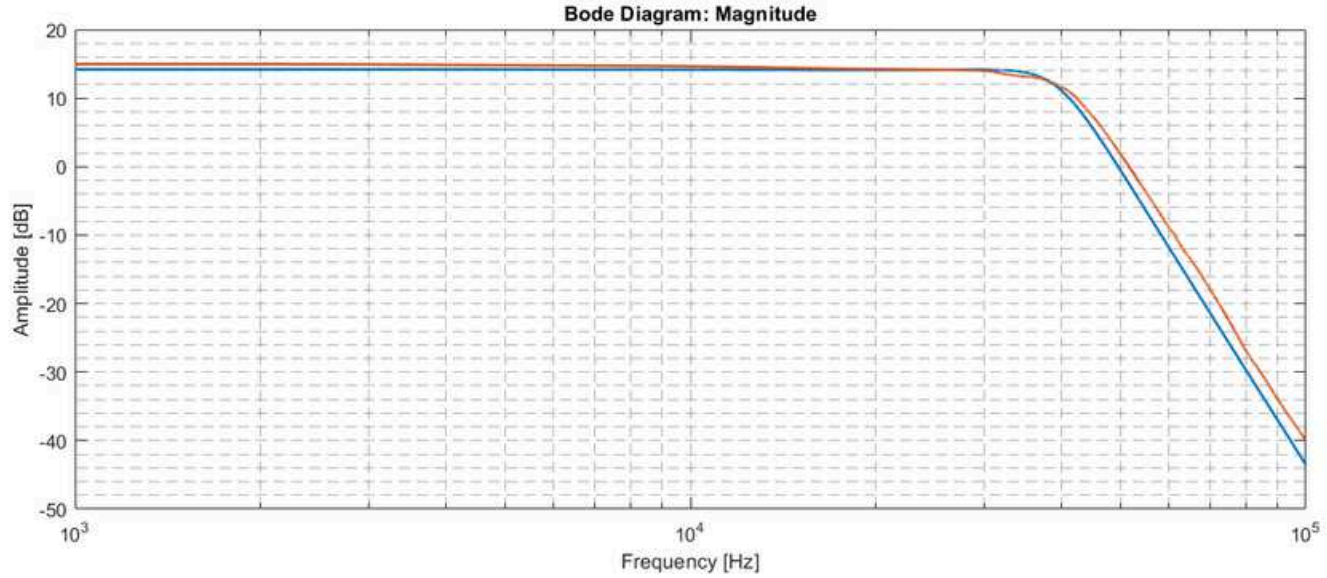
40 kHz Input: As the frequency increases, the filter begins to show its characteristic response: the output signal (blue) exhibits a clear phase lag (−48.96°) and a noticeable reduction in amplitude compared to the input, indicating proximity to the cutoff region.

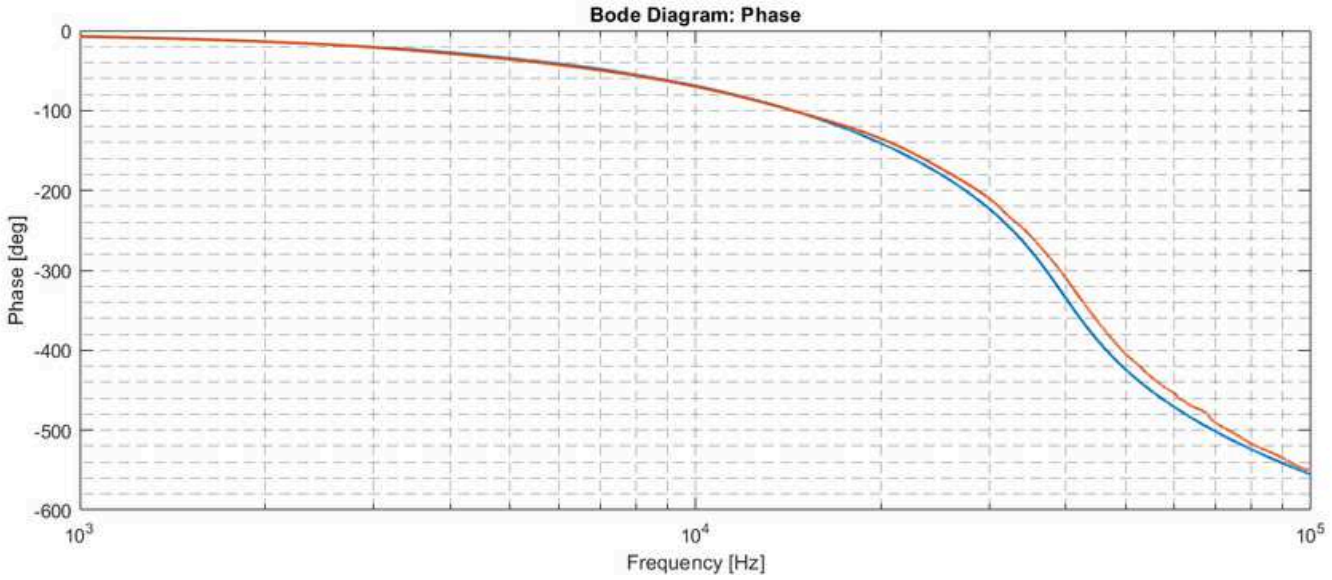
- **Oscilloscope Pictures**



60 kHz Input: At this higher frequency, the filter successfully demonstrates significant attenuation of the signal amplitude and a phase shift approaching −94.39°, confirming effective stopband rejection consistent with a well-functioning low-pass filter.

- **Frequency Characterization**

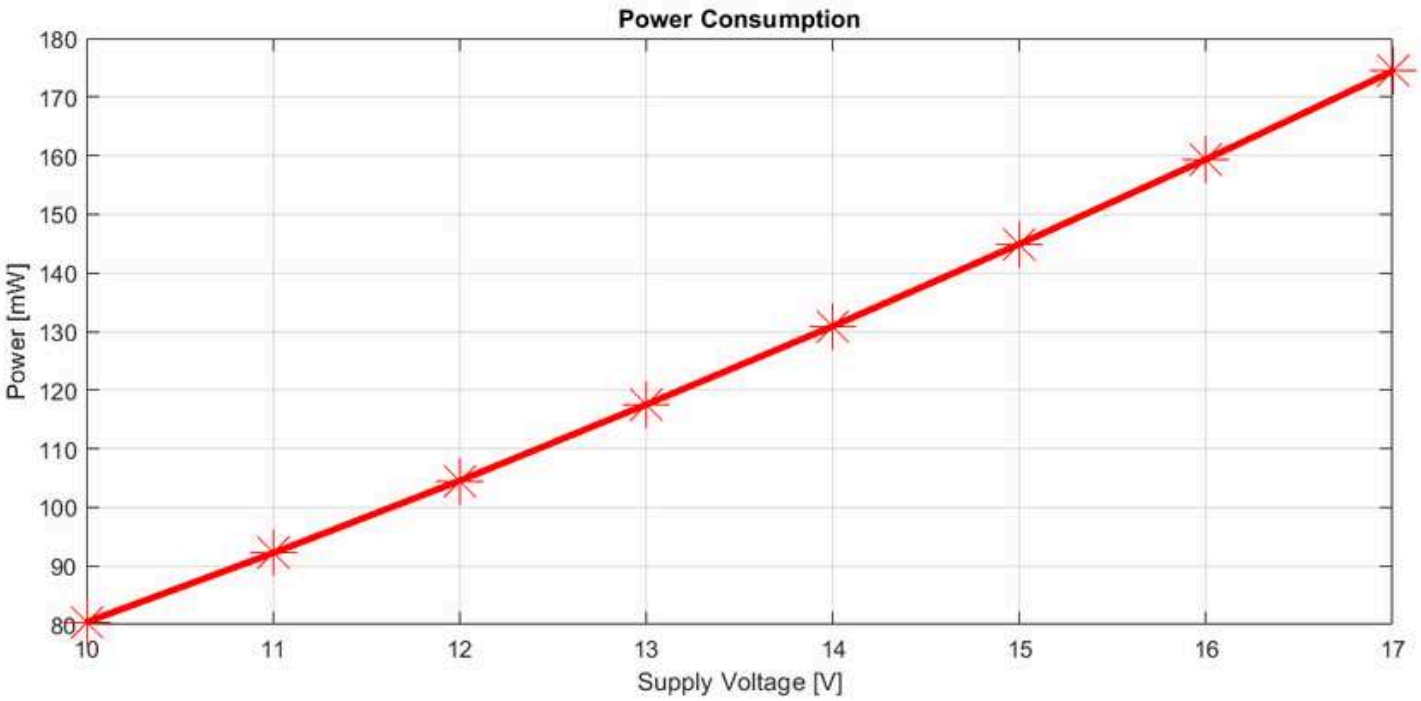




Bode Diagram: The bode diagrams of the experimental measurments, plotted in orange, are in excellent agreement with the simulated response, plotted in blue. Both the magnitude and the phase curves show a similar trend across the entire frequency range, this confirm the validity of the theoretical model used to design the filter.

Frequency	Magnitude (Sim / Tests)	Phase (Sim / Tests)
10 kHz	14.19 / 14.96	-6.87 / -7.5
40k Hz	11.17 / 11.65	-335.27 / -310 .10
61 kHz	-12 .95 / -9.77	-474.81 / -459.98

- **Power Characterization**



Power Consumption: The supply voltage was varied from 10 V, where the filter consumes 80.4 mW, to 17 V, where the filter consumes 174.42 mW. Between those two values the power consumption grows monotonically. The results demonstrate that the filter operates correctly across the entire supply voltage with a power consumption that remains contained within reasonable limits.

Conclusion

Taking everything into account, the results obtained from both the simulations and the practical implementation demonstrate that the designed filter is **capable of effectively attenuating** undesired frequency components with a high degree of accuracy with respect to the design requirements.

The seventh-order structure provides a **steep roll-off** and **improved selectivity**, allowing the filter to achieve the required frequency-response characteristic.
Although high-order filters are generally more sensitive to component tolerances and parameter variations, the Monte Carlo simulations indicate that the proposed design maintains a **good performance** even with the realistic component deviations.

As for the power consumption, it can be seen that the filter shows a relatively **low power requirements**. This is noteworthy considering the complexity of the design, as it demonstrates that the filter is capable of achiving the required frequency-response characteristics while maintaining a **good energy efficiency**.

Furthermore, a comparison between the theoretical predictions, simulation results, and experimental measurements shows a **consistency among them**. The practical implementation closely follows the expected behavior derived from the theoretical analysis, **confirming the validity** of the design methodology and the reliability of the adopted model.